

Niharika Chowdary Mullangi

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(Open to relocate)

SUMMARY

Data Analyst with 4+ years of experience in **Retail, FMCG**, and **Cloud Analytics**, building end-to-end data **platforms, ETL pipelines**, and **ML solutions** to drive actionable business insights. **Actively seeking opportunities** to leverage analytics and AI expertise to enhance decision-making and operational efficiency.

AWARDS & CERTIFICATIONS

- [Microsoft Certified: Azure Data Engineer Associate](#)
- DNA Insta Award(x2): Awarded for leading POS data migration and optimizing downstream performance across reporting workflows. ([2023](#), [2022](#))
- RISE Gracias Award(x2): Recognized for reliable delivery, stakeholder alignment, and on-time milestones across concurrent workstreams. ([2023](#), [2022](#))
- Client & Internal Commendations: Frequent commendations for rapid turnaround on production tasks and proactive issue resolution.

SKILLS

- **Databases:** SQL (Postgres, Snowflake, Microsoft SQL Server, MySQL), NoSQL (Cassandra, MongoDB)
- **Programming Languages:** Python (NumPy, Pandas, Matplotlib, Scikit-learn, PyTorch), R, SQL, PySpark, C++, C#
- **Big Data Ecosystem & ETL:** Apache Spark, Hadoop, Hive, Kafka, Airflow, MapReduce, SSIS, DBT, Data Lake Optimization (partitioning, indexing, views), Pipeline Monitoring & Validation, Informatica, Turbine
- **Cloud Technologies:** Azure (Synapse, Data Factory, Data Lake Storage, Databricks), AWS (Redshift, S3, Glue, Lambda, CloudWatch)
- **Data Visualization:** Power BI, Tableau, Tableau Prep Builder, Looker Studio, Qlik Sense, Excel
- **Machine Learning:** Supervised ML (Linear and Logistic Regression, Decision Trees, RF, Xgboost), Unsupervised ML (K-means, PCA), NLP, Classification techniques (binary classification, multi-class classification), A/B testing
- **Additional Skills:** Git, CI/CD, Shell Scripting, Agile (Scrum, Jira, Confluence)

PROFESSIONAL EXPERIENCE

UNILEVER

Remote, United States

Data Analyst

Aug 2024 - Present

Built a modern Azure-based retail data platform that automated pipelines, powered churn prediction, and accelerated reporting cycles. We built a lightweight platform that consolidates data from POS, loyalty, and e-commerce into one place, enabling retail teams to quickly identify shoppers who might churn and take action.

- Built a unified data platform integrating POS, loyalty, and e-commerce streams using **Python, SQL**, and **Azure Databricks**, enabling real-time visibility into customer engagement and churn risk.
- Developed optimized **SQL** queries and stored procedures across Postgres and Snowflake for reporting and analytics, improving query performance and reducing dashboard refresh time by 35%.
- Developed KPI-driven **Power BI** dashboards with DAX measures and drill-through capabilities, enabling leadership to track revenue and campaign performance.

- Automated feature engineering workflows using **Pandas** and **NumPy**, built machine learning models in **Scikit-learn**, and visualized results with **Matplotlib** to streamline churn prediction analysis.
- Developed **machine-learning** pipelines with scikit-learn logistic regression to predict churn likelihood and identify high-risk segments, supporting targeted retention strategies that improved customer re-engagement.
- Designed interactive **Tableau** dashboards (filters, slicers, parameters), visualizing sales trends, campaign lift, and churn scores across regions, **cutting ad-hoc reporting requests by 40%**.
- Used **Azure OpenAI** to generate human-reviewed personalized offers and campaign copy, reducing marketing turnaround time and improving response rates.
- Created data pipelines in **Azure Data Factory (ADF)** using Linked Services, Datasets, and Pipelines to extract, transform, and load data from multiple sources, including **Azure SQL**, Blob Storage, and Azure Synapse (Data Warehouse), ensuring seamless bidirectional data flow across systems.
- Designed and implemented modern data solutions that supported visualization and reporting layers, enabling faster insight generation and reducing dependency on manual data preparation.
- Implemented Delta Lake with SCD Type 2 and CDC/CDF logic for incremental updates and history tracking, **accelerating dashboard refresh cycles by 60%**.
- Conducted impact analysis and system design reviews to assess how new ETL deployments affected existing business workflows and downstream applications, ensuring zero disruption to ongoing reporting processes.
- Collaborated with **BI** developers and data analysts to align the data model structure with visualization needs in Tableau, ensuring consistent metrics and faster refresh cycles.
- Optimized data ingestion through parallel processing, partitioning, and incremental loads, **reducing load times by 30%** and improving the overall reliability of batch jobs.

INFOSYS

Hyderabad, India

Specialist Programmer (Data Engineer)

Aug 2021 - Jul 2023

Client: P&G

P&G aimed to move its retail data from slow, on-prem systems to a smart cloud-based data platform where every brand team could access reliable, real-time insights. Our team's mission was to make data cleaner, faster, and easier to analyze.

Project 1: On-premises to Azure Delta Lake migration

The goal was to modernize legacy data systems, which were slow and hard to maintain, and move to a unified cloud environment for faster access, more reliable, better governance, and accessible for analytics and machine learning.

- Built automated **ETL pipelines** in Databricks (**PySpark**) and **Azure Data Factory** to bring in data from APIs, databases, and files into ADLS Gen2, **cutting loading time by 75 %**.
- Designed a structured Lakehouse using medallion architecture with fact, dimension, and staging layers, so teams could easily access trusted data for reports and AI models.
- Implemented partition-overwrite logic to **avoid duplicate data** and **speed up processing by 28%**.
- Set up automatic alerts and monitors in Azure to track job health, achieving **99.4%** uptime.
- Used **Excel** functions (**VLOOKUP, INDEX-MATCH, PivotTables**) for early data validation before automating testing with **PySpark** scripts.
- Built an AI-based anomaly detection script to catch schema and data errors before they caused failures, **lowering data inconsistencies by 30%**.

Project 2: Loyalty & planogram data onboarding

Once the base system was stable, the next challenge was bringing in loyalty and planogram data, huge datasets that help P&G understand customer behavior and store layouts. The goal was to make these insights faster and more accurate.

- Created incremental data pipelines in ADF and Databricks to handle 40+ million records, making sure both daily and historical data stayed consistent.
- Built fact, dimension, and aggregate tables using **SQL DDL scripts** and **CTEs** for easier trend analysis.
- Optimized data models and created views and stored procedures, **improving dashboard performance by 70%**.

- Designed dashboards in **Power BI** with **slicers, filters**, and **DAX** measures that visualized promotions, store performance, and customer loyalty trends.
- Built optimized SQL views and materialized summaries for faster BI retrieval and consistent KPIs.
- Automated schema, join, and null checks with PySpark validation modules, reducing manual QA effort by **30 %**.
- Conducted **A/B testing** on promotions to see what worked best, helping marketing teams make better campaign decisions and improving ROI by **15%**.

Project 3: Change requests & retailer-specific enhancements

As the system grew, retailers requested new hierarchies and data granularity. We also had to make pipelines faster, more stable, and easier to maintain.

- Enhanced existing pipelines to include new retailer data and product hierarchies while keeping schema consistency.
- Partnered with clients/SMEs to capture migration and data-mapping rules; wrote impact analyses for upcoming CRs/new builds.
- Optimized transformations and queries using indexing, partitioning, clustering, and broadcast joins, **reducing query runtime by up to 40 %**.
- Tuned Spark jobs with predicate pushdown and adaptive execution, **achieving 40 % faster runtimes**.
- Implemented **CI/CD** pipelines in **Azure DevOps**, so updates could be deployed safely without breaking production jobs.
- Built a LangChain-based assistant that converted natural language questions into **SQL queries**, allowing analysts to get answers faster without technical help.
- Mentored **20+** engineers and standardized **PySpark** and **SQL** coding templates across the team.

Project 4: Data warehouse modernization (implementing slowly changing dimensions)

After stabilizing all major data flows, the next goal was to make the system smarter, capable of tracking every data change automatically and delivering near real-time updates. This phase modernized everything

- Migrated **50+** legacy systems from Parquet to Delta Lake, enabling SCD Type 2 (historical tracking) and CDC (change detection).
- Added Change Data Feed (CDF) to send only updated data downstream, reducing refresh time by 60% and redundant transfers by 80%.
- Built an AI-based metadata summarizer using LangChain + OpenAI API to automatically document job runs and pipeline changes.
- Orchestrated Airflow DAGs to manage created triggers and schedules, track/insert metadata in Postgres, and send automated job alerts for failure or delay events.
- Enhanced historical analytics with feature-engineered fields from Delta data, improving forecast precision by 30 %.
- Strengthened compliance with RBAC, Key Vault, managed identities, and PII masking to meet GDPR and enterprise data protection standards.

TSTRANSCO

Hyderabad, India

Internship

- Utilized **SQL** and **Excel** to analyze energy transmission performance data, developing trend reports and outage frequency insights that supported infrastructure upgrade planning.
- Designed executive-ready dashboards in Power BI/Excel to present power system performance metrics, enabling stakeholders to identify reliability gaps and prioritize investment decisions.

RASHTRIYA ISPAT NIGAM LIMITED

Visakhapatnam, India

Internship

- Assisted in collecting, cleaning, and organizing operational datasets to support process optimization initiatives across production lines.
- Gained hands-on experience in analyzing large datasets using **Excel** and **SQL**, contributing to reporting tasks and efficiency tracking for plant operations.

Internship

- Assisted in mapping signal operation metrics into a centralized Access database, supporting automation of track monitoring workflows.
- Reviewed historical system failure logs and built summary reports to support reliability initiatives across multiple stations.

PROJECTS

Online Course Enrolment Analysis

- Analyzed over 500,000 course enrollment records from a university dataset using PostgreSQL to identify key trends in student preferences, drop rates, and course completion.
- Implemented a columnar data warehouse layer to handle analytical queries efficiently, improving performance and scalability for large datasets.
- Optimized Cassandra partitions and automated reporting in PostgreSQL, accelerating query performance by 35%.

Global Life Expectancy Analysis

- Built an interactive Tableau dashboard visualizing global disparities in life expectancy using public datasets from the WHO and World Bank, covering over 150 countries.
- Performed extensive data cleaning, transformation, and feature engineering in Python (Pandas, NumPy) to prepare metrics such as GDP per capita, healthcare expenditure, and literacy rates for analysis.

Food Establishment Inspections Dashboard

- Developed a **star-schema dimensional** model using **ER Studio** and scripted schema implementation via **DDL**
- Used **Talend** for staging, transformation, and surrogate key management to ensure data integrity
- Developed dashboards visualizing violation counts, risk levels, and inspection frequency trends

Budget Pulse Web Application

- Developed “Budget Pulse,” a financial management web application to help users track expenses, manage budgets, and forecast spending, offering an intuitive and user-friendly experience for effective personal finance planning.
- Engineered and deployed the application using C#, .NET MVC, Razor Pages, and Entity Framework, integrating a scalable PostgreSQL database hosted on Azure, with robust features including secure authentication, predictive budget forecasting based on historical trends, and automated CI/CD pipelines via GitHub Actions.

IMDB Sentiment Analysis

- Built a scalable sentiment analysis pipeline using PySpark and NLP techniques to classify 50,000 IMDb movie reviews as positive or negative, delivering actionable insights for content success prediction in the entertainment industry.
- Trained sentiment classification models using engineered NLP features (tokenization, lemmatization, stop word removal, TF-IDF), achieving 89% accuracy, and identified high-impact terms influencing sentiment classification outcomes.

EDUCATION

UNIVERSITY OF SOUTH FLORIDA

Master of Science in Artificial Intelligence & Business Analytics (GPA: 4/4)

Tampa, FL

Aug 2023 - May 2025

NATIONAL INSTITUTE OF TECHNOLOGY

Bachelor of Technology in Electrical and Electronics Engineering (GPA: 3.68/4)

India

Aug 2017- Apr 2021